Predictable Early, Translated Late: A Layer–Time Lens for Diffusion Image Editors

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Abstract

Between reading an edit instruction and painting pixels, what does a diffusion-transformer (DiT) image editor represent? We build a layer–time lens that decodes intermediate residuals through the model's frozen output head and pairs readouts with linear probes and activation interventions (ablation, transplant, steering). On fixed underspecified recoloring prompts, the *realized* outcome is predictable from early layers while the frozen head still reads noise (car color: 0.667 balanced accuracy vs. chance 0.333, p=0.002). That target-image code is *translated late* into the velocity field near layers 52–54 of 60: the crossover midpoint changes by under one layer between tested timesteps, although its width is timestep-dependent. The boundary replicates in an independent finetune and a four-step distilled checkpoint of the same Qwen lineage; held-out early best-of-five selection validates the readout on a second architecture. Transplant establishes sufficiency; in the tested recolor setting, steering shows writable hue requires a spatial carrier. Truncating classifier-free guidance after step two saves 45% of forward passes on tested edits.

Introduction

Between reading "turn the car red" and painting it, what does a diffusion image editor actually do? The mechanics of diffusion suggest a gradual-development story, each of sixty blocks and twenty denoising steps contributing its small share – but this paper measures that story and finds it wrong. In the model we study, the realized edit is *predictable early* from an internal target-image code, and only *translated late*, in the last few blocks, into the flow field that does the painting.

Prior work provides partial readouts of generative internals. The logit and tuned lenses project transformer residuals through a final or learned output map (nostalgebraist 2020; Belrose et al. 2023). In diffusion, Diffusion Lens interprets text-encoder states; DiffLens and ICM probe or steer denoiser features; and DreamReader and sparse autoencoders provide broader extraction and feature-analysis tools (Toker et al. 2024; Shi et al. 2025; Zaleska et al. 2026; Prakash et al. 2026; Surkov et al. 2025). Linear-representation work motivates a probe-readable basis, while multimodal reasoners externalize visual thoughts as sketches or whiteboards (Park, Choe, and Veitch 2024; Hu et al. 2024; Menon, Zemel, and Vondrick 2024; Wu et al. 2024; Li et al. 2025); we instead test for an unrendered internal image, complementing the broader thinking-with-images taxonomy (Su, Xia et al. 2025). Recent work also detects early cross-seed convergence, predicts final quality from early activations, or allocates adaptive image-editing budgets (Kwon, Lee, and Choi 2026; Guo, Wei, and Jing 2026; Qu et al. 2026). None jointly decodes an instruction-based DiT's image stream across both transformer depth and denoising time while separating explicit-prompt readability from the realized outcome under a fixed prompt.

A second line studies how to produce edits or intervene on hidden states. Modern editors build on diffusion, latent/flow formulations, and guidance (Ho, Jain, and Abbeel 2020; Song, Meng, and Ermon 2021; Rombach et al. 2022; Lipman et al. 2023; Liu, Gong, and Liu 2023; Ho and Salimans 2022). InstructPix2Pix and Prompt-to-Prompt learn or manipulate diffusion editing controls (Brooks, Holynski, and Efros 2023; Hertz et al. 2023); ROME and activation-patching practice transfer hidden states, while activation engineering adds steering directions (Meng et al. 2022; Heimersheim and Nanda 2024; Turner et al. 2023). Concept Lancet already uses representation transplant for diffusion editing (Luo et al. 2025). Recent analyses further show that detectable diffusion features can fail under steering and that multi-mediator patching can hide interaction effects (Cassano et al. 2026; Vaidyanathan et al. 2026). We therefore use transplant only as a sufficiency test and steering only as a bounded writability test, not as a standalone editing method or a necessity proof.

We build a layer–time lens for a production DiT editor, Qwen-Image-Edit-2511 (60 blocks) (Peebles and Xie 2023; Qwen Team 2025), by decoding intermediate residual streams through the model's own frozen output head, and – crucially – pair passive readings with linear probes and controlled activation interventions (ablation, transplant, steering) that test what those activations can install or write.

Naively porting the logit-lens recipe reproduces the familiar mid-stack "nothing happens yet" appearance – and that appearance is itself misleading. With source and an underspecified recoloring instruction fixed across 48 seeds, a generation-level probe predicts the color ultimately chosen

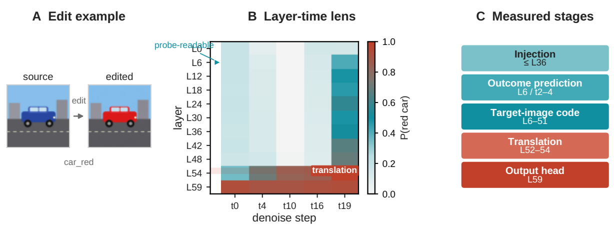

Figure 1: **Predictable early, translated late. A**: source and realized edit. **B**: frozen-head $P(\text{red car})$ across layer and step; the outcome is probe-readable at L6, but translation into the head-readable velocity convention occurs at L52–54. **C**: the five measured stages used throughout the paper.

from layer 6, where the frozen head still reads noise. The outcome is held in a basis the head cannot parse until a narrow band near the top translates it into the velocity field that paints the result.

The main contributions of this work are as follows:

- **An instrument.** A layer–time instrument for a DiT image editor, combining a training-free frozen-head lens with tuned and differential variants, paired with linear probes and three intervention primitives (ablate, transplant, steer) with stated write semantics, under a falsification-first protocol – pre-registered predictions, per-image visual verdicts, and bit-exact gates.
- **A mechanism: the editor thinks with an internal image.** The realized edit is predictable early under a fixed underspecified instruction, carried as a target-image code, and translated into the velocity code near layers 52–54. Across the two densely measured timesteps, the crossover midpoint differs by under one layer while its width varies from one to four layers; the boundary also replicates in a same-lineage fine-tune and a 4-step distillation. Transplant establishes sufficiency; in the tested recolor setting, steering and a rank ladder show that readable attributes require a spatial carrier and a variable-dependent write subspace.
- **Validation and a lever.** Held-out early best-of-five selection validates the readout on Qwen and an independent architecture, with about 68% measured compute reduction. Separately, truncating classifier-free guidance after step 2 saves 45% of forward passes across tested edit families; dropping CFG entirely produces failures consistent with a non-redundant role in scope resolution rather than outcome selection.

This is exploratory, single-model-family work; every verdict was made on the images themselves, and we are explicit about where the lens is unreliable.

Proposed Method: A Layer–Time Lens and Intervention Probes

Figure 2 summarizes the full pipeline; here we specify each component. We study Qwen-Image-Edit-2511, a 60-block DiT editor whose image stream concatenates noisy target tokens with always-clean condition tokens. All quantities below are read from the *post-block residual* of selected blocks via forward hooks, captured in the model's native `bfloat16`: image-stream activations reach absmax $\sim 3\times 10^9$ even at layer 0, so a cast to `float16` silently overflows most elements to `inf` (bf16 capture is a lossless copy, not a downcast).

Notation and layer–time readout. Let $h^{(i)}_{\ell,t} \in \mathbb{R}^d$ denote the post-block residual after block $\ell \in \{0,\dots,59\}$ at denoising step t, for image-stream row i (target patch, condition patch, or a hooked subset); $h^{\text{tar}}_{\ell,t}$ stacks the target-patch rows used for all image-space readouts below (condition rows are captured only for explicitly stated controls). A lens operator $\mathcal{L}_{\ell,t} : \mathbb{R}^{n\times d} \to \mathbb{R}^{n\times 64}$ maps these residuals to a target-grid velocity estimate $\tilde{v}_{\ell,t}$. Logit- and Jacobian-style lenses in language models align internal states with output units along depth and token position; a DiT patch has no vocabulary, so we define a *layer–time lens* as a map from (ℓ, t) to an image and then a score on a fixed grid:

$$I_{\ell,t} = \mathcal{V}\big(\Pi_t\big(\mathcal{L}_{\ell,t}(h^{\text{tar}}_{\ell,t})\big)\big)\,, \qquad \Pi_t(\tilde{v}) = x_t - \sigma_t \tilde{v}, \quad (1)$$

where $\mathcal{V}$ is the frozen VAE decoder. We score a fixed edit-region crop of $I_{\ell,t}$ with CLIP ViT-L/14 (Radford et al. 2021). With unit-normalized image and prompt-ensembled text embeddings e_I, e_T, and CLIP's learned logit scale γ, the forced-choice probability over the reported label bank $\{y_k\}$ is

$$[\mathcal{M}_{\ell,t}]_k \;=\; \operatorname{softmax}_k\Big(\gamma\, e_T(y_k)^\top e_I(\operatorname{crop}(I_{\ell,t}))\Big). \quad (2)$$

Figures 1 and 2 instantiate (1)–(2): the image-editor analogue of lexical alignment grids in logit/Jacobian-lens visualizations, not a full $\partial \mathbf{y}/\partial h$ (a donor-scalar Jacobian basis is tested in the supplement and does not coincide with probe or write directions).

Lens operators. The *time lens* uses the terminal head output at step t:

$$\hat{x}_0(x_t, t) = x_t - \sigma_t\, v_\theta(x_t, t), \quad (3)$$

decoded through the VAE—what the model currently believes it will output. The *depth lens* asks the same question at intermediate depth by re-invoking the model's frozen output head (non-affine `AdaLayerNormContinuous` + linear `proj_out`, bit-exact whether invoked at layer 59 or earlier):

$$v_{\ell,t} = \operatorname{Head}(h^{\text{tar}}_{\ell,t}), \qquad \hat{x}_{0,\ell,t} = x_t - \sigma_t\, v_{\ell,t}, \quad (4)$$

then $I_{\ell,t} = \mathcal{V}(\hat{x}_{0,\ell,t})$. Because `norm_out` has no learned scale, rescaling $h^{\text{tar}}_{\ell,t}$ to the terminal norm is a no-op (raw vs. rescaled agree to `max_rel_diff`$=0$); we use the raw head everywhere with a standing self-consistency gate on every grid. The *tuned lens* (Belrose et al. 2023) fits a per-(ℓ,t) ridge map from d=3072 hidden dimensions to the 64-d conditional-pass velocity, one dictionary per cell,

$$\hat{v}_{\ell,t} = W_{\ell,t} h^{\text{tar}}_{\ell,t} + b_{\ell,t}, \quad W_{\ell,t} = V^\top H (H^\top H + \lambda I)^{-1}, \quad (5)$$

after token-wise centering; λ is chosen per cell on a held-out 10% *token* split, distinct from zero-shot tests that hold out

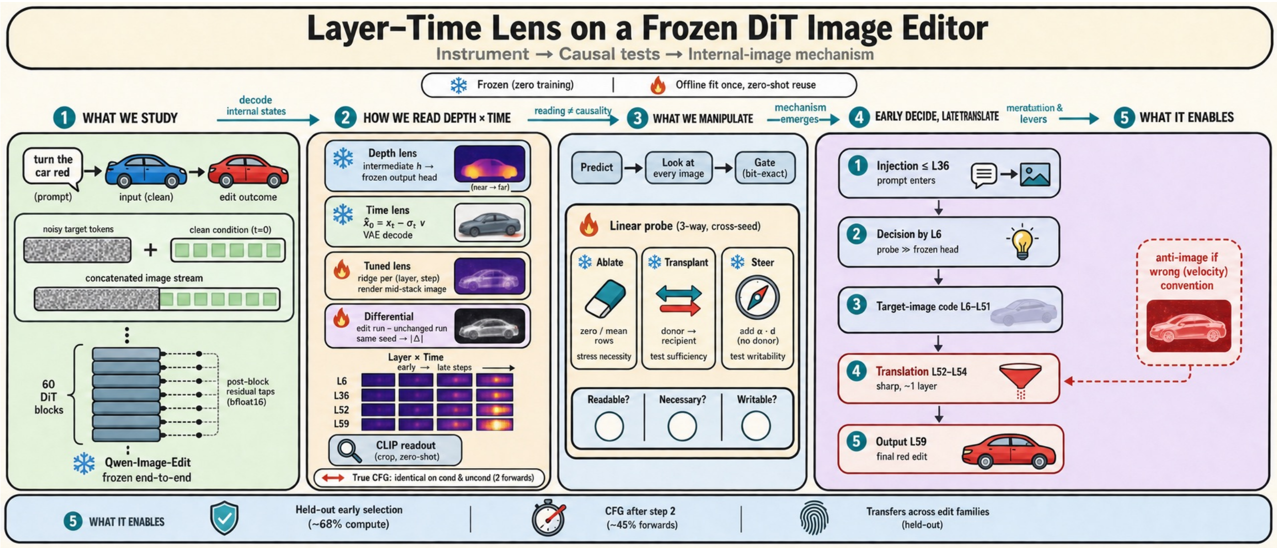

Figure 2: **Framework overview.** Frozen Qwen-Image-Edit-2511 on a concatenated noisy-target / clean-condition image stream with post-block residual hooks (`bfloat16`). *Read*: depth, time, tuned, and differential lenses decode previews, then crop-level CLIP scoring on matched true-CFG passes. *Test*: linear probes plus ablate, transplant, and steer interventions under predict–look–gate protocol. *Mechanism* (schematic): early decision and target-image code, late L52–L54 translation into the velocity field; bottom: held-out validation and training-free levers discussed in text.

an entire edit family and source image (hyperparameters in the supplement). The *differential lens* differences edit and control runs that share a seed (step-0 latents are then bit-identical):

$$D_{\ell,t} = \left| I_{\ell,t}^{\mathrm{edit}} - I_{\ell,t}^{\mathrm{ctrl}} \right|, \tag{6}$$

isolating the internal edit plan when the unchanged prompt is a faithful no-op; a control-fidelity gate on correlation with the source flags cases where it is not.

Linear probes. Neither (2) nor a lens image is itself a causal claim. We additionally fit a lightweight probe $p(y \mid h_{\ell,t}) = \mathrm{softmax}(W h_{\ell,t})$ on hidden states (cross-seed evaluation), reading the same (ℓ, t) grid in label space rather than pixel space—the basis mismatch against the frozen head in (4) is central to our mechanism results.

Intervention probes. Three forward-hook overwrites of the post-block residual test distinct properties, applied identically to both CFG passes:

$$\begin{aligned} h &\leftarrow \mathbf{0} \text{ or } \bar{h} && \text{(ablate: necessity stress test),} \\ h^{(\mathrm{rec})} &\leftarrow h^{(\mathrm{don})} && \text{(transplant: sufficiency),} \\ h &\leftarrow h + \alpha d && \text{(steer: linear writability),} \end{aligned} \tag{7}$$

with α in units of per-row RMS and d a probe direction; an empty ablation spec reproduces the baseline bit-for-bit. Repeated ablation can drive activations off-manifold, so we treat it only as a coarse stress test.

Protocol: predict, look, gate. Three rules make the *reasoning* reproducible, not just the runs. **Predict, then run**: every intervention was pre-registered; refutations are reported as prominently as confirmations, and several headline results are pre-registered predictions that failed informatively. **Look at every image**: no verdict rests on CLIP alone – every run carries a per-image visual verdict recorded before the aggregates were read, and where the two disagree the image wins. **Gate the instrument**: the empty-ablation and raw-vs-rescaled checks are standing regressions re-run per experiment. Porting the method to another editor needs only a native-dtype residual hook, the frozen-head gate, and region-restricted probes – the recipe executed unchanged in the cross-model replication below.

Experiments

Experimental Setup

We evaluate on Qwen-Image-Edit-2511, a 60-block DiT editor, on edit batteries spanning five categories – color, background, style, removal, and addition – over natural photographs and one flat-art source (10 cases for the time-lens saturation sweep and the preview lever, 20 for the IoU-based onset and CFG-truncation checks); every quantitative number is paired with a per-image visual verdict recorded before the aggregate is read, per the predict-look-gate protocol above. Three readouts do the reporting: the frozen output head's decode, scored zero-shot with CLIP against a small label bank; cross-seed probes, including a 48-seed fixed-prompt outcome probe whose independent unit is one generation; and, for the tuned lens, held-out R^2 against the true conditional-pass velocity. Two further checkpoints sharing Qwen-Image-Edit-2511's transformer classes – FireRed-Image-Edit-1.1, an independent fine-tune, and Rapid-AIO, a 4-step distilled merge – let us ask whether a finding persists beyond one training run (tested below). Statistical units follow the claim: explicit-prompt patch probes are basis diagnostics, whereas fixed-prompt tests contribute one mean-pooled vector per generation. Their p-values use fold-matched label permutations; held-out selection uses exact Poisson-binomial inference (Boogu) or exact within-group rank tests (Qwen).

The Realized Edit Is Predictable Early

Question. Does the edit unfold gradually, or is the final outcome readable before the frozen head agrees?

In time. If the edit surfaced progressively, the readout would saturate late and at roughly the same rate. Running the time lens over a battery of 10 cases across 5 edit categories, the step at which $P(\text{target})$ first reaches within 0.05 of its final value is instead *edit-dependent* and mostly *immediate*: color, background, and style edits reach the criterion at step 0–2, and only a wholesale new-object insertion (*add_boat*) does so as late as step 7. A finer IoU-mask-based onset measurement over 20 cases shows the split is categorical, not continuous: 18 of 20 – spanning final-edit-area fraction 0.003 to 0.99 – onset at step 0 regardless of how large or novel-looking the finished edit is, and no continuous proxy predicts the delay (edge-novelty $\rho{=}0.24$, $p{=}0.30$; area $\rho{=}{-0.44}$, running the *wrong* way). Only the two edits that insert genuinely novel structure with no existing pixels to build on – *add_boat* and *add_runner* – start late.

In depth, the naive lens says "late". The frozen depth lens tells a different-looking story. Decoding *car_red*'s intermediate layers through the model's own head, the edit becomes head-readable only in a sharp snap between layers 48 and 54 of 60 – exactly the familiar mid-stack "nothing happens yet" appearance of an LLM logit lens.

But the realized outcome is predictable 40 layers earlier. An explicit-prompt probe first reveals the basis mismatch: predicting which of three colors was requested reaches 0.883 patch-token accuracy at **layer 6**, step 4, where the frozen head reads only $P(\text{red car}){=}0.111$ (Figure 3). To separate outcome prediction from reading a color word, we hold both source and the underspecified prompt "change the car to a different color" fixed across 48 new seeds and label each final image post hoc. One mean-pooled vector per generation predicts the realized color at L6/step 4 with 0.667 balanced accuracy (chance 0.333; $p{=}0.002$). Thus the outcome, not merely the instruction, is linearly predictable early; this does not claim irreversible commitment.

Prediction generalizes and supports selection. Table 1 summarizes continuous-hue replications and held-out best-of-five tests. Natural-photo mug and tree edits replicate while the heterochromatic-eyes task is null ($p{=}0.796$); the independent 40-block Boogu editor also carries an early hue readout, although its terminal layer is stronger. Selection succeeds on Boogu and two Qwen target tasks. Actual interrupted captures plus winner reruns save about 68% versus full best-of-five, but only select the closest available candidate.

Where the instruction enters. A complementary text-stream ablation locates where the prompt is *consumed*: zeroing the model's text tokens in only the first 11 blocks garbles the whole scene, while zeroing them after layer 36 barely

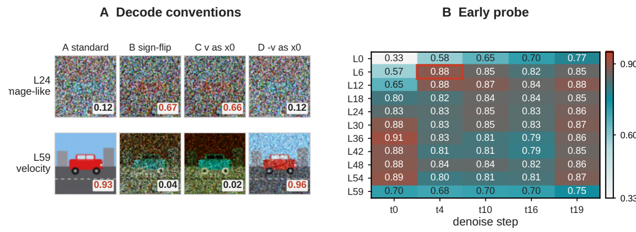

Figure 3: **Decode convention flips; the probe reads early.** **A**: four *car_red* decodes reverse convention between L24 and L59. **B**: a requested-color basis diagnostic reaches 0.883 patch-token accuracy at L6/$t4$ while the frozen head still reads noise (chance 0.33). Patches are not independent experimental units; the generation-level fixed-prompt result is 0.667 over 48 generations.

Table 1: Early prediction and held-out selection. Q/B denote Qwen/Boogu; parentheses give chance or random-selection baselines.

Setting	Readout	Metric	Result	Evidence
Car fixed	Q L6/$t4$	Bal. acc.↑	.667 (.333)	p=.002
Car hue	Q L6/$t2$	MAE↓	22.3→8.3°	R^2=.815
Tree hue	Q L6	MAE↓	55.9→11.4°	p=.005
Mug hue	Q L6	MAE↓	17.0→10.7°	p=.005
Boogu hue	B L4/$t2$	MAE↓	39.9→6.9°	p=.005
Boogu select	B L4 best-5	Success↑	8/12 (20%)	$p{=}4.1{\times}10^{-5}$
Tree select	Q L6/$t2$	MAE↓	95.2→46.5°	$p{=}3.7{\times}10^{-7}$
Car select	Q best-5	MAE↓	25.7→11.1°	p=.0086

dents the edit ($P(\text{red}){=}0.993$). The instruction is fused into the image stream early; for this color edit, later text tokens are dispensable by layer 36 – the "injection" stage of the circuit in Figure 1, whose full text-band table is in the supplement. Together with the probe, this gives the paper's five-stage picture: injection (≤L36), early outcome prediction, a target-image code, translation (L52–54), and the output head. **Takeaway.** Time onset is mostly immediate; the frozen depth lens looks late, but generation-level probes read the realized outcome from L6.

The Edit Is Translated Late: Two Codes

Question. If probes read the outcome early, what mid-stack code is converted at L52–54? Three measurements answer this: at an intermediate layer the frozen head yields a per-token vector v_ℓ we normally treat as a velocity, forming $\hat{x}_0 = x_t - \sigma_t v_\ell$ (variant A). Decoding v_ℓ instead *directly*, as if it were already a clean latent (variant C), inverts the story: for *car_red* at step 4, at L24 the standard decode A scores $P(\text{red}){=}0.12$ (still wrong-colored) while the direct decode C already scores 0.66; at L59 this reverses (A $= 0.93$, C $= 0.02$). The mid-stack is not blank – it encodes the target content in an *image-like* convention, and a narrow band near the top reconciles it to the velocity convention the head consumes. (This is a whole-vector reweighting, not a literal sign flip: essentially no individual channel of v_ℓ has a negative slope against v_{59}, mean 0.003.)

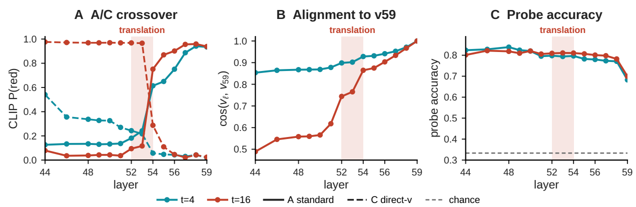

Figure 4: **Translation near L52–54.** Dense L44–L59 sweep on *car_red*. **A–B**: the decode-crossover midpoint changes by under one layer between $t{=}4, 16$, while transition width changes from four layers to one. **C**: probe accuracy stays flat through L58 and drops only at L59.

The conversion location is stable; its width is timestep-dependent. A dense every-layer sweep from L44 to L59 (Figure 4) locates the A/C crossover precisely: transition midpoint L53.6 at step 16 (width one layer) and L52.7 at step 4 (width four layers). The midpoints differ by under one layer, so the boundary's *location* is stable across these timesteps even though its sharpness is not. Crucially, the cross-seed probe accuracy is close to *flat* (0.77–0.84) across every layer from L44 through L58 and drops only at the terminal layer L59 (0.68–0.70). So the translation that changes what the head can read (L52–54) and the loss of linear separability (L59) are two *distinct* events sharing a neighborhood: the target-image code survives the translation intact and is degraded only by the final, velocity-specific projection. Coarse family sweeps place measurable style and background crossovers in the same terminal band; removal and addition remain inconclusive because their between-variant CLIP margins are too small.

A tuned lens renders the image-like code, layer by layer. To test whether the mid-stack code is image-like rather than a probe artifact, we fit a per-(layer, timestep) tuned lens, as described above, on exactly three source images – one each from *remove*/*background*/*addition* – and evaluate the entire *car_red* source and color family *zero-shot*. Held-out R^2 rises with depth from 0.73–0.89 at L6 to 0.998 at L59, whose frozen-head decode reproduces the true velocity bit-for-bit. Every tested depth down to L6 yields a coherent scene, refuting a pre-registered readability boundary: L52–54 marks the frozen head's projection, not the representation. At step 0 the pixels are pure noise, yet Figure 5 shows the blue car at L24, orange-red at L36, and fully red at L52. The dictionaries transfer without retraining to six further held-out edits; *add_runner* also reproduces delayed onset internally.

The anti-image, mechanized. This also explains a striking artifact: the mid-stack *frozen*-head decode at later steps is a color-*inverted* ghost (a cyan car under a red-brown sky) – the image-like mid-stack code forced through a head built to consume a velocity, since $\hat{x}_0 = x_t - \sigma v$ flips the sign of an image-like quantity. The tuned lens at the same cell is already coherent. **Takeaway.** Mid-stack states are image-like; the L52–54 band reconciles them to the velocity convention the head consumes.

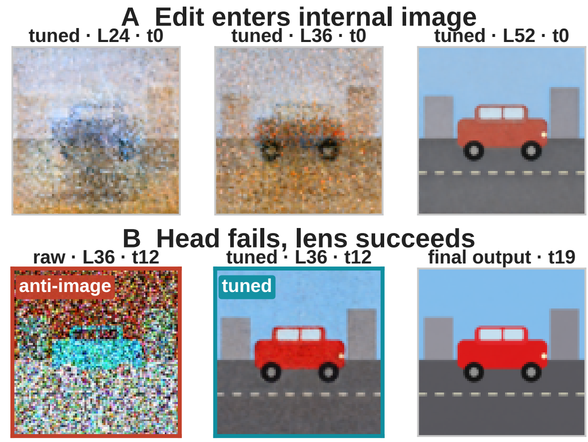

Figure 5: **Internal image vs. frozen-head ghost. A**: the zero-shot tuned lens shows the edit enter a coherent internal image at $t{=}0$. **B**: for the same L36/t12 state, the frozen head yields an anti-image while the tuned lens recovers the edited scene.

Transplantable, Writable, and Carrier-Bound

Reading a lens shows only that a representation is *consistent with* an edit, not what it can install. Whole-token ablation across many layers drives the edited region off-manifold, so we use it only as a coarse stress test rather than evidence for fine-grained necessity.

Transplant establishes sufficiency. Transplanting a *car_red* donor's car-region hidden states into a *car_green* recipient (which alone scores $P(\text{red}){=}0.011$) flips the output to red from *any* of four donor bands, including **L48–59** (0.940). The donor representation is therefore sufficient to install the edit. A content control confirms specificity: splicing the same donor's *background*-region tokens (right band, wrong content) yields a garbled "blue car" (0.991), not red.

Readability does not imply unconstrained writability. A single stored *direction* – the probe's d_{red}, no donor pass – added to the recipient's car rows gives a sharp sigmoidal dose-response (0.024 at $\alpha{=}0.5$, 0.977 at $\alpha{=}1$, saturating to $\alpha{=}8$); a random direction of equal norm is inert (0.017), and the negated direction overshoots to blue (0.945), reproducing the anti-image under an active intervention. But writability is bounded: at effective doses the steered region is not a recolored car but a flat red patch that *replaces* the car's geometry – the direction installs *which* hue, not *where to stop*. Supplying that missing "where" with the edited object's own silhouette mask turns every previously-failing dose into a clean in-place recolor (Figure 6). And conjuring an *object* outruns any single direction: the identity probe direction is inert (the read and write subspaces differ), while a rank ladder over the donor−recipient difference resolves identity coarse-to-fine – category at rank 1, a coherent instance by rank 4, donor-faithful by rank 64 (2% of the model dimension). The read and write subspaces differ, and a writable attribute still requires a spatial carrier.

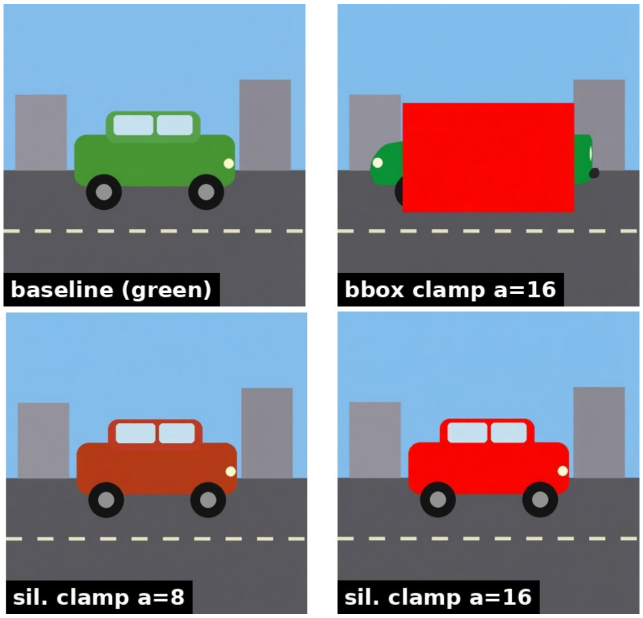

Figure 6: **A spatial carrier restores clean writability.** Steering red inside a bounding box eventually overwrites the car with a flat patch (top); the same direction restricted to the car silhouette preserves geometry across effective doses (bottom).

Rank controls isolate the object write basis. On one fixed 61-patch lake hull, projecting only the donor–recipient difference's top singular modes gives a coarse-to-fine identity ladder: rank 1 already recruits the boat category but produces an incoherent cluster; rank 4 produces one coherent but wrong instance; rank 16 recovers the donor's wooden material; and rank 64 (2% of model width, 89% of difference energy) approaches the full donor transplant. A same-dimensional random rank-64 basis is inert. Thus the ladder reflects alignment to a donor-derived write subspace, not merely intervention rank, and explains why a linearly readable identity probe need not be directly writable.

Coherent interventions conserve attention routing. Why does a full-state transplant preserve the car while direction steering overwrites it? Following a reference-matched routing audit used for visual-token reduction (Cho et al. 2026), a side recorder recomputes conditional-pass joint attention without changing the model output, over five seeds, five layers, five steps, three query groups, and four key groups. Relative to the natural red-edit reference, the content-matched transplant changes routing by only 0.0031 total variation and preserves a coherent car in 5/5 seeds (Table 2). An equal-size background transplant is as distorted as destructive zero/mean overwrite and fails in 5/5. Red-direction steering writes red in 5/5 but also has high routing distortion and produces the flat patch in Figure 6. Across seeds, the most distorted coherent condition remains below the least distorted incoherent one in 5/5 (mean gap 0.0430, exact one-sided sign-flip $p=0.031$). This supports routing conservation as an explanation for the tested coherent transplant; it is not a necessity or architecture-general claim.

Table 2: Attention-routing audit on the car intervention. TV is distortion from the natural red-edit reference (↓); coherence is a blinded image-level verdict over five seeds.

Intervention	Routing TV ↓	Coherent
Car-state transplant	.0031	5/5
Red-direction steer	.0546	0/5
Zero overwrite	.0620	0/5
Mean overwrite	.0668	0/5
Background transplant	.0677	0/5

A Preview Lever: The Differential Lens at 15% NFE

The differential lens (*Proposed Method*) isolates the internal edit plan from the reconstruction. It first settles a caveat left open above: token-level R^2 *inside* the edit region is 0.88–0.97, matching the full-image value, so the tuned dictionaries carry the edit-specific signal itself, not incidental scene detail.

Content is early; separation is gradual. A pre-registered prediction that D would already localize tightly at every step is refuted, informatively. The edit *content* is present at step 0 (the tuned lens already shows red eyes, a black trunk), but the *differential* at step 0 is a diffuse whole-frame field – the edit prompt perturbs the entire velocity relative to the no-op, not just the edited region – and condenses onto the edit's footprint only gradually (localization of D's top mass at L52 climbs monotonically across steps, e.g. $0.21 \rightarrow 0.47 \rightarrow 0.67 \rightarrow 0.83$ for a cat's-eyes recolor). *Present* (content in the internal image) and *separated* (the differential has found where it lives) are different quantities. The no-op control also caught one flat-art scene that *cannot* "do nothing" (it renders a transparency checkerboard, and is discarded), motivating the control-fidelity gate applied before any differential number is trusted.

A preview lever. Because the differential carries the edit's footprint, we ask whether it does so early enough to be useful. At step 2 – the third of 20, ≈15% of the forward passes – across a 10-case, five-family battery of natural photographs (all passing the control-fidelity gate, $r \in [0.86, 0.99]$), the median IoU between $D_{\text{L52},t2}$'s top mass and a final-output difference mask is 0.33, $6.6\times$ chance, and all 10 cases sharpen from step 0 to step 2 (Figure 7): an object slated for removal lights up as a ghost, a keep-vs-replace background segmentation emerges unsupervised, and a tiny recolor glows at its exact site. Because the no-op control is *independent of the edit prompt*, it is computed once per source image and cached; previewing a *candidate* edit then costs three forward passes plus one frozen-dictionary decode, not a full 20-step run – a training-free way to triage many prompts against one image before running any to completion.

Generalization Across Checkpoints

Most mechanistic findings so far concern one model lineage. We re-ran the depth-lens crossover and cross-seed probe

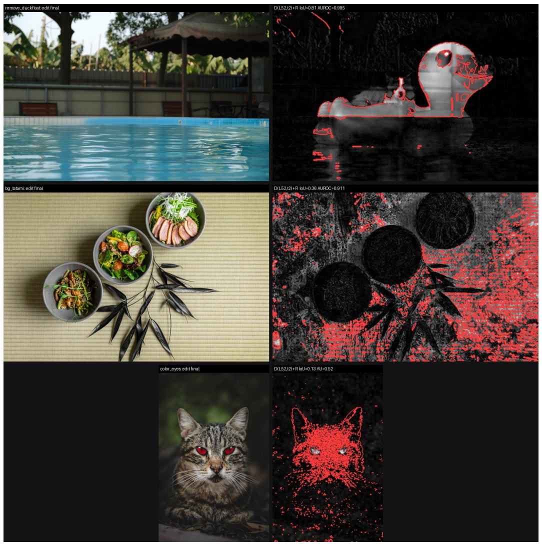

Figure 7: **Differential preview at 15% NFE.** Final/preview pairs for removal, background replacement, and recolor after three of twenty steps; red contours mark the final difference mask.

on the two further checkpoints from *Setup* (FireRed-Image-Edit-1.1 and Rapid-AIO). **The translation boundary and explicit-prompt early readability replicate** (Table 3). FireRed has the same flat-then-dip probe shape, and Rapid-AIO preserves the boundary under a four-step schedule. The readable direction also transfers between Qwen and FireRed, rather than merely recurring at the same layer index. The full tuned lens ports to FireRed as well (deep-layer R^2 0.97–0.999; anti-image reproduced).

The readout basis transfers, not only the boundary. A three-rung ladder decodes FireRed with the Qwen translator dictionary frozen (L0), with only FireRed calibration substituted (L1), or with a fully native refit (L2). Across a six-case held-out battery spanning five edit families, all 108 deep cells (six cases $\times$ L52/56/59 $\times$ six steps) order L0 $<$ L1 $<$ L2; recalibration recovers a median 89% of the frozen-to-native gap, while the fully frozen dictionary never falls below $R^2{=}0.86$. Even *style_ink*, a family absent from translator fitting, has a frozen floor of 0.885. Thus the deep readout basis is reusable within this model lineage, although its calibration remains checkpoint-specific; this does not test an architecturally unrelated DiT.

The differential law and preview lever transfer. On two held-out FireRed photos that pass the unchanged-control gate ($r{=}0.933/0.990$), L52 localization rises with denoising for both tree recolor ($0.90{\to}0.97$) and runner addition ($0.29{\to}0.84$). Their $t{=}2$ preview IoUs are 0.676 and 0.217, above the registered 0.15 threshold and 0.05 chance, and both improve over $t{=}0$ ($0.610/0.142$). Early content, gradual separation, and the three-step preview therefore reproduce on this fine-tuned checkpoint.

Table 3: Cross-checkpoint boundary/probe transfer (Qwen$\leftrightarrow$FireRed); L12/t4 direction cosine is .833.

Checkpoint	A/C band	Probe: early $\to$ L59	Transfer
Qwen	L52–54	.883 @ L6/t4 $\to$ .68–.70	source
FireRed	L53–56	.852 @ L6/t4 $\to$ .69–.70	.840/.917
Rapid-AIO	L52–54	–	–

Noise level, not schedule index, keys the dictionary. At the Rapid step where fractional-position and σ matching differ, σ matching wins all six deep cells (median $+0.055$ R^2). Within σ coverage, the frozen Qwen readout reaches $R^2{=}0.927$–0.995 and ties a Rapid-native refit in 13/24 cells; only the uncovered terminal $\sigma{=}0.02$ fails, and a matched entry recovers it. Thus the dictionary is keyed to noise level, not sampler position.

A Training-Free Lever: Truncating Classifier-Free Guidance

Running the unconditional pass only for the first two of 20 steps leaves all tested families intact at a **45% reduction in forward passes**; a 20-case battery confirms it (mean 1.77$\times$ speedup). With no CFG, recolor, removal, and addition still land, but background replacement composites around unerased source structure (P(beach)$=0.45$ vs. ≥ 0.996), consistent with a non-redundant role in scope resolution. A four-step distilled checkpoint also lands 20/20 cases (7.7$\times$ faster), preserving the exposed structure while compressing the sampling trajectory.

Limitations

Translation/intervention evidence uses Qwen plus two same-lineage checkpoints; Boogu supports only outcome prediction/selection. The fixed-prompt probe has one synthetic source and two natural-image replications; one eye case is unsuitable. Selection has 12 groups per task and cannot create absent targets. One-way L6 hue transfer (mug$\to$tree $p{=}0.002$; reverse $p{=}0.968$) precludes an object-invariant subspace. Because CLIP can prefer a flat patch to a true recolor, quantitative claims carry pre-registered visual verdicts. The dense boundary and carrier study center on the flat-art car; its silhouette comes from a source–edit difference, not automatic segmentation. The fixed top-10% preview mask can over-expand tiny edits, so IoU is a localization diagnostic, not segmentation ground truth. We claim tested predictability, transplant sufficiency, and carrier-bounded writability, not commitment or necessity.

Conclusion

Outcomes are predictable early; Qwen-lineage image codes translate to velocity late. The lenses separate readability, sufficiency, and carrier-bounded writability without claiming commitment, necessity, or universality.

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